

Part 1 (No AI used, only HI)

Q1) Prove $\vec{a} = [a_1 \ a_2 \ a_3]^T$ $\vec{b} = [b_1 \ b_2 \ b_3]^T$ $\vec{c} = [c_1 \ c_2 \ c_3]^T$

a) $a \cdot (b \times c) = (a \times b) \cdot c$

LHS = $a \cdot (b \times c)$

$$b \times c = \begin{bmatrix} b_2 c_3 - b_3 c_2 \\ b_3 c_1 - b_1 c_3 \\ b_1 c_2 - b_2 c_1 \end{bmatrix}$$

$$\therefore a \cdot (b \times c) = \begin{pmatrix} a_1 (b_2 c_3 - b_3 c_2) \\ + a_2 (b_3 c_1 - b_1 c_3) \\ + a_3 (b_1 c_2 - b_2 c_1) \end{pmatrix} = \begin{pmatrix} a_1 b_2 c_3 - a_1 b_3 c_2 \\ + a_2 b_3 c_1 - a_2 b_1 c_3 \\ + a_3 b_1 c_2 - a_3 b_2 c_1 \end{pmatrix}$$

$$\therefore a \cdot (b \times c) = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_2 b_1 c_3 - a_3 b_2 c_1$$

RHS = $(a \times b) \cdot c$

$$a \times b = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

$$\therefore (a \times b) \cdot c = \begin{pmatrix} (a_2 b_3 - a_3 b_2) c_1 \\ + (a_3 b_1 - a_1 b_3) c_2 \\ + (a_1 b_2 - a_2 b_1) c_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 c_1 - a_3 b_2 c_1 \\ + a_3 b_1 c_2 - a_1 b_3 c_2 \\ + a_1 b_2 c_3 - a_2 b_1 c_3 \end{pmatrix}$$

$$\therefore (a \times b) \cdot c = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_2 b_1 c_3 - a_3 b_2 c_1$$

$\therefore$ As ① = ② $\therefore$ LHS = RHS

Hence Proved

$$b) \ a \times (b \times c) = (a \cdot c) b - (a \cdot b) c$$

$$LHS = a \times (b \times c)$$

$$b \times c = \begin{bmatrix} b_2 c_3 - b_3 c_2 \\ b_3 c_1 - b_1 c_3 \\ b_1 c_2 - b_2 c_1 \end{bmatrix}$$

$$a \times (b \times c) = \begin{bmatrix} a_2 b_1 c_2 - a_2 b_2 c_1 - a_3 b_3 c_1 + a_3 b_1 c_3 \\ a_3 b_2 c_3 - a_3 b_3 c_2 - a_1 b_1 c_2 + a_1 b_2 c_1 \\ a_1 b_3 c_1 - a_1 b_1 c_3 - a_2 b_2 c_3 + a_2 b_3 c_2 \end{bmatrix} \quad \text{--- (i)}$$

$$RHS = (a \cdot c) b - (a \cdot b) c$$

$$a \cdot c = a_1 c_1 + a_2 c_2 + a_3 c_3 \quad a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$(a \cdot c) b = \begin{bmatrix} a_1 c_1 b_1 + a_2 c_2 b_1 + a_3 c_3 b_1 \\ a_1 c_1 b_2 + a_2 c_2 b_2 + a_3 c_3 b_2 \\ a_1 c_1 b_3 + a_2 c_2 b_3 + a_3 c_3 b_3 \end{bmatrix}$$

$$(a \cdot b) c = \begin{bmatrix} a_1 b_1 c_1 + a_2 b_2 c_1 + a_3 b_3 c_1 \\ a_1 b_1 c_2 + a_2 b_2 c_2 + a_3 b_3 c_2 \\ a_1 b_1 c_3 + a_2 b_2 c_3 + a_3 b_3 c_3 \end{bmatrix}$$

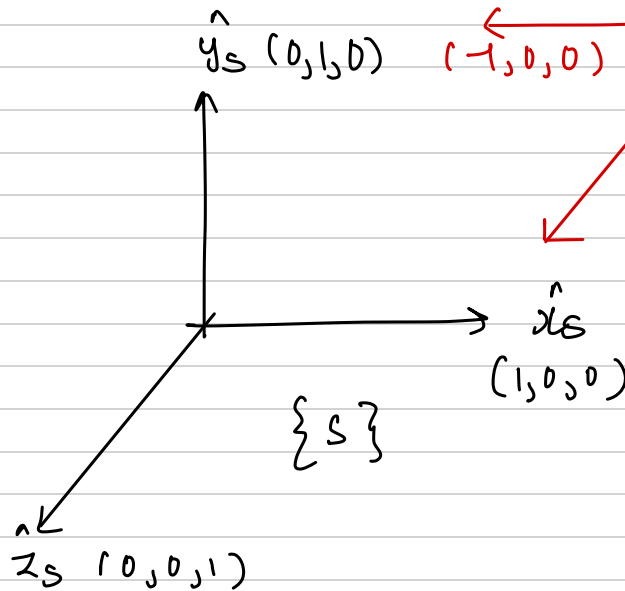
$$\therefore (a \cdot c) b - (a \cdot b) c = \begin{bmatrix} a_2 b_1 c_2 + a_3 b_1 c_3 - a_2 b_2 c_1 - a_3 b_3 c_1 \\ a_1 b_2 c_1 + a_3 b_2 c_3 - a_1 b_1 c_2 - a_3 b_3 c_2 \\ a_1 b_3 c_1 + a_2 b_3 c_2 - a_1 b_1 c_3 - a_2 b_2 c_3 \end{bmatrix} \quad \text{--- (ii)}$$

$$\therefore \text{(i)} = \text{(ii)} \quad \therefore LHS = RHS$$

Hence Proved

Q2)

a)



Property: $\hat{z}_a = \hat{x}_a \times \hat{y}_a$
 $\hat{z}_b = \hat{x}_b \times \hat{y}_b$

Assumption:

$$R_{ab} = R_b^a$$

b) Rotation matrices

We know $R_1^0 = R_{01} = \begin{bmatrix} x_1 \cdot x_0 & y_1 \cdot x_0 & z_1 \cdot x_0 \\ x_1 \cdot y_0 & y_1 \cdot y_0 & z_1 \cdot y_0 \\ x_1 \cdot z_0 & y_1 \cdot z_0 & z_1 \cdot z_0 \end{bmatrix}$

$$\therefore R_{sa} = \begin{bmatrix} x_a \cdot x_s & y_a \cdot x_s & z_a \cdot x_s \\ x_a \cdot y_s & y_a \cdot y_s & z_a \cdot y_s \\ x_a \cdot z_s & y_a \cdot z_s & z_a \cdot z_s \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\therefore R_{sb} = \begin{bmatrix} x_b \cdot x_s & y_b \cdot x_s & z_b \cdot x_s \\ x_b \cdot y_s & y_b \cdot y_s & z_b \cdot y_s \\ x_b \cdot z_s & y_b \cdot z_s & z_b \cdot z_s \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

c) $R_{sb} \rightarrow$ rotation matrix $\in SO(3)$

property $R \in SO(3) \Rightarrow R^T = R^{-1}$

$$\therefore R_{sb}^{-1} = R_{sb}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \left(\text{calculated without using matrix inverse} \right)$$

Basically from drawing

$$R_{sb} \cdot R_{sb}^{-1} = I \quad \therefore R_{sb}^{-1} = R_{bs} \quad (s \rightarrow b)$$

$$\therefore R_{sb}^{-1} = R_{bs} = \begin{bmatrix} x_s \cdot x_b & y_s \cdot x_b & z_s \cdot x_b \\ x_s \cdot y_b & y_s \cdot y_b & z_s \cdot y_b \\ x_s \cdot z_b & y_s \cdot z_b & z_s \cdot z_b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Hence verified (by drawing and coordinates)

d) we know $R_1^0 R_2^1 = R_2^0$

Also, $R_{01} R_{12} = R_{02}$ (same)

Here, $R_{sa} R_{ab} = R_{sb}$

$$\therefore R_{ab} = R_{sa}^{-1} R_{sb} = R_{sa}^T R_{sb}$$

$$R_{ab} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(computed again without using matrix inv.)

from drawing,

$$R_{ab} = R_b^a = \begin{bmatrix} x_b \cdot x_a & y_b \cdot x_a & z_b \cdot x_a \\ x_b \cdot y_a & y_b \cdot y_a & z_b \cdot y_a \\ x_b \cdot z_a & y_b \cdot z_a & z_b \cdot z_a \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$\therefore$ Hence verified (by drawing and coordinates)

e) $R = R_{sb} = R_{x_b}^{-90^\circ} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-90^\circ) & -\sin(-90^\circ) \\ 0 & \sin(-90^\circ) & \cos(-90^\circ) \end{bmatrix}$

$$\therefore R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_1 = R_{sa} R = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_2 = R R_{sa} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now let us analyse.

$R_{sa} \rightarrow$ orientation $R \rightarrow$ rotation. abt x by -90°

in $R_1 \rightarrow R$ is multiplied after R_{sa} , which means R is rotating the x axis after R_{sa} has taken place. Hence the x axis here is the body axis (after R_{sa})

but, in $R_2 \rightarrow R$ is multiplied before R_{sa} , this means that the x axis which rotates by -90° has not yet seen the action of R_{sa} hence it is the fixed axis (before R_{sa})
(world)

$\therefore R_1 \rightarrow$ rotⁿ abt body-fixed x axis.

$\therefore R_2 \rightarrow$ rotⁿ abt world-fixed x axis.

f) $p_b = (1, 2, 3)$

$$\therefore \vec{p}_s = R_{sb} \vec{p}_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (1, 3, -2)$$

g) $p_s = (1, 2, 3)$

$$p' = R_{sb} \vec{p}_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (1, 3, -2)$$

$$p'' = R_{sb}^T p_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (1, -3, 2)$$

here $R_{sb}^T = R_{bs}$ = rotn matrix $s \rightarrow b$

$\therefore p'' \rightarrow$ changing coordinates from frame $\{s\}$ to $\{b\}$ without moving the point.

But, R_{sb} = rotn matrix $b \rightarrow s$

$\therefore p' = R_{sb} \vec{p}_s$ does not change the coordinates, but instead rotates $\vec{p}_s$ in the same frame $\{s\}$.

h) $\vec{w}_s = (3, 2, 1)$

$$\vec{w}_a = R_{as} \vec{w}_s = R_{sa}^T \vec{w}_s = R_{sa}^T \vec{w}_s$$

$$\therefore \vec{w}_a = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = (1, -3, -2)$$

Q3) $\vec{p} = (\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{2}})$ wrt fixed frame $\hat{x}_0, \hat{y}_0, \hat{z}_0$

$\rightarrow$ First rotn: abt fixed x by $30^\circ \Rightarrow R_{\hat{x}, 30^\circ}$

$$R_{\hat{x}, 30^\circ} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c(30^\circ) & -s(30^\circ) \\ 0 & +s(30^\circ) & c(30^\circ) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & -1/2 \\ 0 & 1/2 & \sqrt{3}/2 \end{bmatrix}$$

$\rightarrow$ Second rotn: abt fixed y by $135^\circ \Rightarrow R_{\hat{y}, 135^\circ}$

$$R_{\hat{y}, 135^\circ} = \begin{bmatrix} c(135^\circ) & 0 & s(135^\circ) \\ 0 & 1 & 0 \\ -s(135^\circ) & 0 & c(135^\circ) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$\rightarrow$ Third rotn: abt fixed z by $-120^\circ \Rightarrow R_{\hat{z}, -120^\circ}$

$$R_{\hat{z}, -120^\circ} = \begin{bmatrix} c(-120^\circ) & -s(-120^\circ) & 0 \\ s(-120^\circ) & c(-120^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

a) $\vec{p}' = R_{\hat{z}, -120^\circ} R_{\hat{y}, 135^\circ} R_{\hat{x}, 30^\circ} \vec{p} \quad \text{--- (1)}$

$$R_{\hat{z}, 120^\circ} R_{\hat{y}, 135^\circ} = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$R_{\hat{z}, 120^\circ} R_{\hat{y}, 135^\circ} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} & \frac{1}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{1}{2} & -\frac{\sqrt{3}}{2\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore R_{\hat{z}, 120^\circ} R_{\hat{y}, 135^\circ} R_{\hat{x}, 30^\circ} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} & \frac{1}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{1}{2} & -\frac{\sqrt{3}}{2\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\therefore R_{\hat{z}, 120^\circ} R_{\hat{y}, 135^\circ} R_{\hat{x}, 30^\circ} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{3}{4} - \frac{1}{4\sqrt{2}} & -\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} & \frac{1}{4} & -\frac{3}{4\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$

Now substitute in ①

$$\vec{p}' = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{3}{4} - \frac{1}{4\sqrt{2}} & -\frac{\sqrt{3}}{4} & -\frac{\sqrt{3}}{4\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} & \frac{1}{4} & -\frac{3}{4\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad \text{--- ①}$$

$$\therefore \vec{p}' = \begin{bmatrix} \frac{1}{2\sqrt{6}} - \frac{3}{4\sqrt{6}} + \frac{1}{4\sqrt{12}} - \frac{\sqrt{3}}{4\sqrt{2}} - \frac{\sqrt{3}}{8} \\ \frac{1}{2\sqrt{2}} + \frac{1}{4\sqrt{2}} + \frac{1}{8} + \frac{1}{4\sqrt{2}} - \frac{3}{8} \\ -\frac{1}{\sqrt{6}} + \frac{1}{2\sqrt{12}} - \frac{\sqrt{3}}{4} \end{bmatrix} = \begin{bmatrix} \frac{-2\sqrt{6}-\sqrt{3}}{12} \\ \frac{2\sqrt{2}-1}{4} \\ \frac{-\sqrt{6}-\sqrt{3}}{6} \end{bmatrix}$$

b) Comparing eqn ① with $\bar{p}' = R \bar{p}$

$$R = R_{\hat{z}, 120^\circ} R_{\hat{y}, 135^\circ} R_{\hat{x}, 30^\circ}$$

$$\therefore R = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{3}{4} - \frac{1}{4\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4\sqrt{2}} & \frac{1}{4} - \frac{3}{4\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$

Q4) $\{a\} \rightarrow (\hat{x}_a, \hat{y}_a, \hat{z}_a)$ $\{b\} \rightarrow (\hat{x}_b, \hat{y}_b, \hat{z}_b)$
 $\{c\} \rightarrow (\hat{x}_c, \hat{y}_c, \hat{z}_c)$ Prove $R_a^b R_b^c = R_a^c$

$$\text{Given, } \begin{aligned} \hat{x}_b^n &= r_{11} \hat{x}_a^n + r_{21} \hat{y}_a^n + r_{31} \hat{z}_a^n \\ \hat{y}_b^n &= r_{12} \hat{x}_a^n + r_{22} \hat{y}_a^n + r_{32} \hat{z}_a^n \\ \hat{z}_b^n &= r_{13} \hat{x}_a^n + r_{23} \hat{y}_a^n + r_{33} \hat{z}_a^n \end{aligned}$$

$$\therefore R_a^b = \begin{bmatrix} \hat{x}_a \cdot \hat{x}_b & \hat{y}_a \cdot \hat{x}_b & \hat{z}_a \cdot \hat{x}_b \\ \hat{x}_a \cdot \hat{y}_b & \hat{y}_a \cdot \hat{y}_b & \hat{z}_a \cdot \hat{y}_b \\ \hat{x}_a \cdot \hat{z}_b & \hat{y}_a \cdot \hat{z}_b & \hat{z}_a \cdot \hat{z}_b \end{bmatrix} = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix}$$

$$\text{Also Given, } \begin{aligned} \hat{x}_c^n &= s_{11} \hat{x}_b^n + s_{21} \hat{y}_b^n + s_{31} \hat{z}_b^n \\ \hat{y}_c^n &= s_{12} \hat{x}_b^n + s_{22} \hat{y}_b^n + s_{32} \hat{z}_b^n \\ \hat{z}_c^n &= s_{13} \hat{x}_b^n + s_{23} \hat{y}_b^n + s_{33} \hat{z}_b^n \end{aligned}$$

$$\therefore R_b^c = \begin{bmatrix} \hat{x}_b \cdot \hat{x}_c & \hat{y}_b \cdot \hat{x}_c & \hat{z}_b \cdot \hat{x}_c \\ \hat{x}_b \cdot \hat{y}_c & \hat{y}_b \cdot \hat{y}_c & \hat{z}_b \cdot \hat{y}_c \\ \hat{x}_b \cdot \hat{z}_c & \hat{y}_b \cdot \hat{z}_c & \hat{z}_b \cdot \hat{z}_c \end{bmatrix} = \begin{bmatrix} s_{11} & s_{21} & s_{31} \\ s_{12} & s_{22} & s_{32} \\ s_{13} & s_{23} & s_{33} \end{bmatrix}$$

$$\text{Now, } R_a^c = \begin{bmatrix} \hat{x}_a \cdot \hat{x}_c & \hat{y}_a \cdot \hat{x}_c & \hat{z}_a \cdot \hat{x}_c \\ \hat{x}_a \cdot \hat{y}_c & \hat{y}_a \cdot \hat{y}_c & \hat{z}_a \cdot \hat{y}_c \\ \hat{x}_a \cdot \hat{z}_c & \hat{y}_a \cdot \hat{z}_c & \hat{z}_a \cdot \hat{z}_c \end{bmatrix}$$

$$x_a \cdot x_c = S_{11} x_a \cdot x_b + S_{21} x_a \cdot y_b + S_{31} x_a \cdot z_b \\ = S_{11} x_{11} + S_{21} x_{12} + S_{31} x_{13}$$

→ Similarly we can fill all 9 elements in R_a^c

$$R_a^c = \begin{bmatrix} S_{11}x_{11} + S_{21}x_{12} + S_{31}x_{13} & S_{11}x_{21} + S_{21}x_{22} + S_{31}x_{23} \\ S_{12}x_{11} + S_{22}x_{12} + S_{32}x_{13} & S_{12}x_{21} + S_{22}x_{22} + S_{32}x_{23} \\ S_{13}x_{11} + S_{23}x_{12} + S_{33}x_{13} & S_{13}x_{21} + S_{23}x_{22} + S_{33}x_{23} \end{bmatrix}$$

→ we can split them into matrix multiplication

$$R_a^c = \underbrace{\begin{bmatrix} S_{11} & S_{21} & S_{31} \\ S_{12} & S_{22} & S_{32} \\ S_{13} & S_{23} & S_{33} \end{bmatrix}}_{R_b^c} \underbrace{\begin{bmatrix} x_{11} & x_{21} & x_{31} \\ x_{12} & x_{22} & x_{32} \\ x_{13} & x_{23} & x_{33} \end{bmatrix}}_{R_a^b}$$

$$\therefore R_a^c = R_b^c R_a^b \neq R_a^b R_b^c$$

↪ correct formula !!

∴ Eqn told to prove is incorrect

$$R_a^c = R_a^b R_b^c \quad \times \quad \Rightarrow \quad R_a^c = R_b^c R_a^b \quad \checkmark$$

Q5) $R \in SO(3)$

a) Prove : $R(v \times w) = R(v) \times R(w)$

$$v, w \in \mathbb{R}^3 \quad \therefore v = [v_1 \ v_2 \ v_3]^T, \quad w = [w_1 \ w_2 \ w_3]^T$$

$$\text{We have } v = v_1 \hat{i}_0 + v_2 \hat{j}_0 + v_3 \hat{k}_0$$

$$\text{and } w = w_1 \hat{i}_0 + w_2 \hat{j}_0 + w_3 \hat{k}_0$$

$$\therefore v \times w = (v_2 w_3 - v_3 w_2) \hat{i}_0 + (v_3 w_1 - v_1 w_3) \hat{j}_0 + (v_1 w_2 - v_2 w_1) \hat{k}_0$$

$$R \rightarrow R \circ R^n \text{ from frame } \{0\} \text{ to } \{1\} \Rightarrow R_0$$

$\rightarrow \hat{i}_1, \hat{j}_1, \hat{k}_1 = \text{basis of frame } \{1\}$

$$\text{Unit vector along axis } \{0\} \Rightarrow \vec{x}_0 = \hat{i}_0 + 0 \hat{j}_0 + 0 \hat{k}_0$$

$$\therefore \text{unit vector along axis } \{1\} \Rightarrow \vec{x}_1 = R \vec{x}_0 = \underline{R \hat{i}_0 = \hat{i}_1}$$

$$\text{Similarly, } R \hat{j}_0 = \hat{j}_1, \quad R \hat{k}_0 = \hat{k}_1$$

$$\therefore R(v \times w) = (v_2 w_3 - v_3 w_2)(R \hat{i}_0) + (v_3 w_1 - v_1 w_3)(R \hat{j}_0) + (v_1 w_2 - v_2 w_1)(R \hat{k}_0)$$

$$= (v_2 w_3 - v_3 w_2) \hat{i}_1 + (v_3 w_1 - v_1 w_3) \hat{j}_1 + (v_1 w_2 - v_2 w_1) \hat{k}_1 \quad \text{--- (I)}$$

$$R(w) = v_1 R \hat{i}_0 + v_2 R \hat{j}_0 + v_3 R \hat{k}_0 = v_1 \hat{i}_1 + v_2 \hat{j}_1 + v_3 \hat{k}_1$$

$$\text{Similarly, } R(w) = w_1 \hat{i}_1 + w_2 \hat{j}_1 + w_3 \hat{k}_1$$

$$\therefore R(v) \times R(w) = (v_2 w_3 - v_3 w_2) \hat{i}_1 + (v_3 w_1 - v_1 w_3) \hat{j}_1 + (v_1 w_2 - v_2 w_1) \hat{k}_1 \quad \text{--- (II)}$$

$$\therefore \text{(I)} = \text{(II)}$$

$$\therefore R(v \times w) = R(v) \times R(w)$$

Hence Proved

b) Prove $R \hat{\bar{\omega}} R^T = (\hat{R \bar{\omega}}) \quad \omega \in \mathbb{R}^3$

→ Take arbitrary vector $\bar{x} \in \mathbb{R}^3$, multiply both side

$$\therefore R \hat{\bar{\omega}} R^T \bar{x} = (\hat{R \bar{\omega}}) \bar{x}$$

$$\text{LHS} = R(\hat{\bar{\omega}} R^T \bar{x}) - \text{①}$$

$$\text{RHS} = (\hat{R \bar{\omega}}) \bar{x}$$

Property: $\hat{a} \bar{b} = \bar{a} \times \bar{b}$

$$\therefore \hat{\bar{\omega}} R^T \bar{x} = \bar{\omega} \times R^T \bar{x} - \text{②}$$

put ② in ① $\therefore \text{LHS} = R(\bar{\omega} \times R^T \bar{x})$

from a) $R(\bar{v} \times \bar{w}) = R(\bar{v}) \times R(\bar{w})$

$$\therefore R(\bar{\omega} \times R^T \bar{x}) = R \bar{\omega} \times R R^T \bar{x} = R \bar{\omega} \times \bar{x}$$

$$\therefore \text{LHS} = R \bar{\omega} \times \bar{x}$$

we know $\bar{a} \times \bar{b} = \hat{\bar{a}} \bar{b}$

$$\rightarrow \therefore \text{LHS} = R \bar{\omega} \times \bar{x} = (\hat{R \bar{\omega}}) \bar{x} = \text{RHS}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence Proved

c) Given $R = [r_1 \ r_2 \ r_3]$, $r_i \in \text{columns of } R$
($i \rightarrow 1 \text{ to } 3$)

as $R \in SO(3)$ $\therefore$ columns of orthogonal with magnitude = 1

$$\hookrightarrow \therefore \det(R) = 1 - \text{②}$$

(Property)

$$\therefore r_2 \times r_3 = r_1 - \text{①}$$

③

To prove. $\det(R) = r_1^T (r_2 \times r_3)$

$$\rightarrow \text{LHS} = \det(R) = 1 - \text{from ②}$$

$$\rightarrow \text{RHS} = r_1^T r_1 = r_1 \cdot r_1 = \|r_1\|^2 = 1 \quad (\text{from ③})$$

from ①

$$\therefore \text{LHS} = \text{RHS} = 1$$

Hence Proved

d) To prove $\mathfrak{so}(3)$ [skew-sym. matrix set] $\Rightarrow$ vector space (3×3)

Note: for $V \rightarrow$ vector space

(i) if $u, v \in V$, then $u + v \in V$

(ii) if $u \in V$, then $\lambda u \in V \quad \forall \lambda = \text{scalar}$.

(iii) $0 \in V$

for $\mathfrak{so}(3) \rightarrow$ vector space All (i), (ii), (iii) should be true.

Assume $A = \begin{bmatrix} 0 & -a_1 & -a_2 \\ a_1 & 0 & -a_3 \\ a_2 & a_3 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 0 & -b_1 & -b_2 \\ b_1 & 0 & -b_3 \\ b_2 & b_3 & 0 \end{bmatrix} \therefore A, B \in \mathfrak{so}(3)$

$\therefore A+B = \begin{bmatrix} 0 & -a_1-b_1 & -a_2-b_2 \\ a_1+b_1 & 0 & -a_3-b_3 \\ a_2+b_2 & a_3+b_3 & 0 \end{bmatrix} \Rightarrow \text{skew symmetric} \therefore A+B \in \mathfrak{so}(3)$

$\therefore$ (i) is true for $\mathfrak{so}(3)$
 $\lambda A = \begin{bmatrix} 0 & -\lambda a_1 & -\lambda a_2 \\ \lambda a_1 & 0 & -\lambda a_3 \\ \lambda a_2 & \lambda a_3 & 0 \end{bmatrix} \Rightarrow \text{skew symmetric} \therefore \lambda A \in \mathfrak{so}(3)$
 $(\lambda \rightarrow \text{scalar})$

$\therefore$ (ii) is true for $\mathfrak{so}(3)$

0 matrix $= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $0 + 0^T = 0 \rightarrow \text{skew symmetric} \therefore 0 \in \mathfrak{so}(3)$

$\therefore$ (iii) is true for $\mathfrak{so}(3)$

$\therefore \mathfrak{so}(3) \rightarrow$ vector space $\therefore$ it satisfies all (i), (ii), (iii)

Hence Proved

Dimension

$A = \begin{bmatrix} 0 & -a_1 & -a_2 \\ a_1 & 0 & -a_3 \\ a_2 & a_3 & 0 \end{bmatrix}$

$\rightarrow$ parameters needed to completely define $A = a_1, a_2, a_3$

$\therefore \dim(\mathfrak{so}(3)) = 3$

Basis (A_1, A_2, A_3)

using these parameters

$A = a_1 A_1 + a_2 A_2 + a_3 A_3$

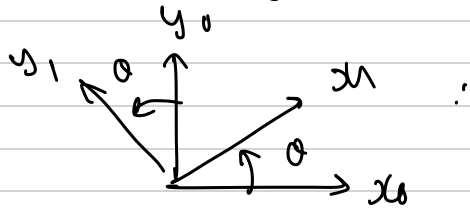
$\therefore A_1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $A_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$A_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Q6) $R \in SO(2)$

→ 2×2 orthogonal matrix, $\det(R) = 1$

a) So basically R is a rotation matrix in 2D plane


$$\therefore R_1^0 = \begin{bmatrix} x_1 \cdot x_0 & y_1 \cdot x_0 \\ x_1 \cdot y_0 & y_1 \cdot y_0 \end{bmatrix}$$
$$\therefore R_1^0 = \begin{bmatrix} \cos \theta & \cos(\theta_0 + \theta) \\ \cos(\theta_0 - \theta) & \sin \theta \end{bmatrix}$$

$$\therefore R = R_1^0 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \in SO(2)$$

($\theta \rightarrow$ parameter)

$$S' \rightarrow \text{unit circle in } \mathbb{R}^2 \therefore S' = \begin{bmatrix} x \\ y \end{bmatrix} \nmid x^2 + y^2 = 1$$

$$\text{Let } x = \cos \theta \therefore y = \sin \theta \Rightarrow S' = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \forall \theta \in \mathbb{R}$$

$$\therefore \text{if we have } S' = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \text{ we can write}$$

$$\text{the equivalent } R \in SO(2) \text{ for it, } R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$\therefore R \in SO(2)$ can be identified by S' Hence Proved

b) When $R \in SO(2)$, $\hat{\omega} = \mathfrak{so}(2) = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}$

$$\text{To prove } R \hat{\omega} R^T = \hat{\omega} \therefore \text{RHS} = \hat{\omega}$$

$$\therefore \text{LHS} = R \hat{\omega} R^T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$
$$= \begin{bmatrix} -\sin \theta \omega & -\cos \theta \omega \\ \cos \theta \omega & -\sin \theta \omega \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$
$$= \begin{bmatrix} -\sin \theta \cos \theta \omega + \sin \theta \cos \theta \omega & -\sin^2 \theta \omega - \cos^2 \theta \omega \\ \sin^2 \theta \omega + \cos^2 \theta \omega & \cos \theta \sin \theta \omega - \cos \theta \sin \theta \omega \end{bmatrix}$$
$$= \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} = \hat{\omega} = \text{RHS} \therefore \text{Hence Proved}$$